

Crime and the Classroom

How Local Violence Affects School Dropout Decisions in Mexico

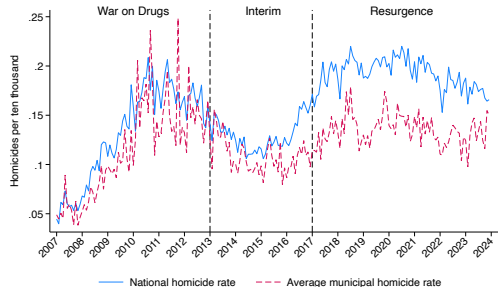
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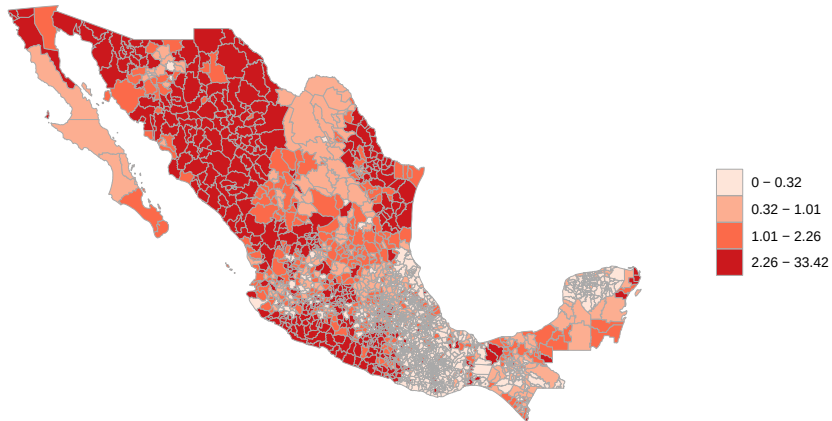
Introduction

Mexico's violence epidemic

- Drug trafficking organizations (DTOs) generate \$19–29 billion USD annually
- Three distinct eras of violence since 2007:
 - **War on Drugs** (2007–2012): Calderón's militarized offensive
 - **Interim Period** (2013–2016): Strategic withdrawal, lower rates
 - **Resurgence** (2017–2024): Historically high, persistent violence
- 2016: Mexico was the world's #2 deadliest conflict zone (behind Syria)
- In 2024: 35,600 persons reported missing



Violence is geographically concentrated



Research question

Does an increase in the local homicide rate cause young people to drop out of school?

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The effect is **theoretically ambiguous**:

Violence $\uparrow$ dropout:

- Fear reduces perceived safety
- Children supplement lost household income
- DTOs recruit youth as enforcers, runners, lookouts

Violence $\downarrow$ dropout:

- Schools function as safe havens
- Higher enforcement risk deters participation
- Social stigma of drug trade discourages recruitment

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$\Rightarrow$ This is an **empirical question**

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Approach:

- Individual-level panel data from ENOE (2007–2024): 841,150 children
- Construct high-frequency monthly dropout indicator
- Instrumental variables: Colombian cocaine seizures $\times$ distance to Pacific coast
- Exploit three eras of violence and enforcement policy

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Preview of findings:

- Pooled IV estimate: small positive effect, imprecise
- **Resurgence period (2017–2024):** large, positive, statistically significant
- Effects concentrated among **high school-aged males**
- Evidence consistent with **DTO recruitment mechanism**

Contributions

1. **Identification:** IV strategy (shift-share) addresses reverse causality and omitted variable bias present in prior TWFE estimates
2. **Sample period:** Covers 2007–2024 — the longest panel in this literature; most prior work covers only 2007–2012
3. **Outcome:** First monthly, individual-level dropout measure for Mexico; avoids institutional reporting incentives in administrative data
4. **New finding:** Evidence of a structural break under AMLO's permissive enforcement regime, consistent with heightened DTO recruitment

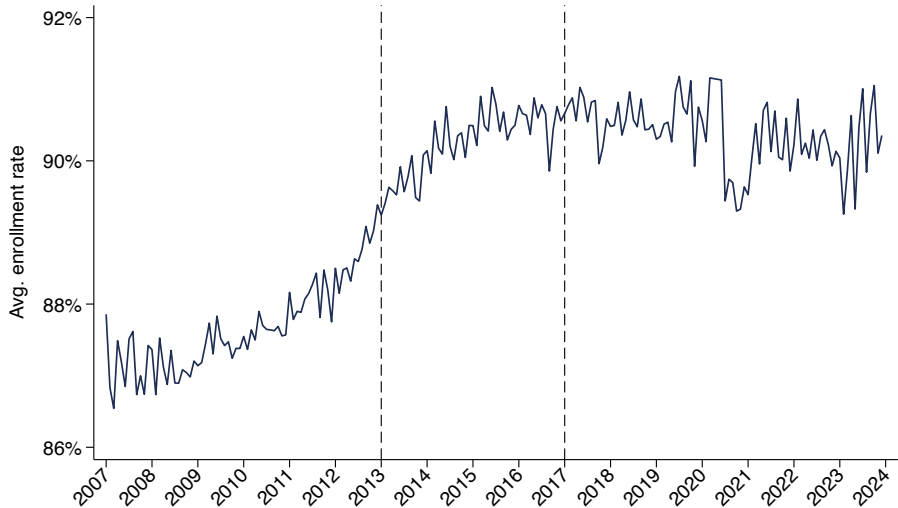
Data

1. ENOE: National Survey of Occupation and Employment
2. Homicides: INEGI official death records
3. Colombian cocaine seizures: Colombian Ministry of Defense
4. Geographic data: INEGI shapefiles

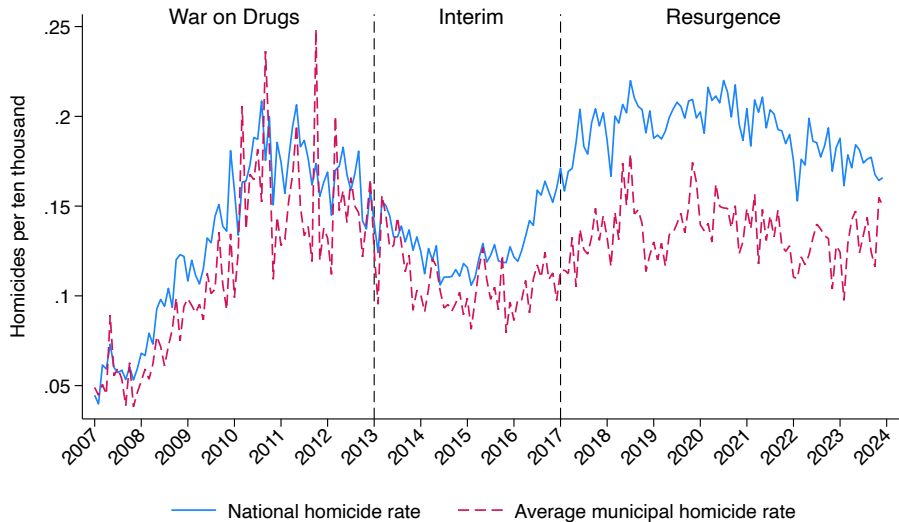
ENOE: rotating individual panel

- Nationally representative rotating panel, 5 consecutive quarters per individual
- Covers 2007–2024; one quarter missing (COVID, Q2 2020)
- Keep only individuals completing all 5 surveys
- **Sample:** 841,150 children aged 12–18 in urban municipalities ($\geq 15,000$ pop.)
- **Dropout:** transition from enrolled $\rightarrow$ not enrolled between two survey waves
 - Sidesteps institutional reporting incentives
- Dropout rates by school level:
 - Secondary: 5.5%
 - High school: 11.1%
- Discontinuity at age 15 = minimum legal working age
- Primary-age children (6–11) excluded:
 - Negligible dropout
 - Theoretical channels don't apply

Enrollment rates have been roughly flat since 2015

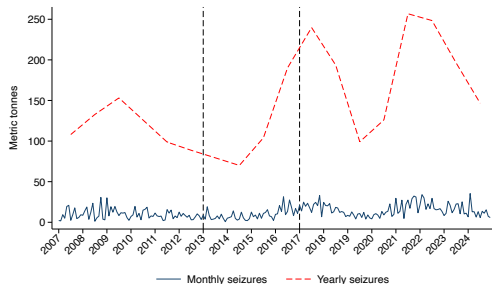


Homicide data: three eras of violence



Colombian cocaine seizures: the supply shock

- Large coastal seizures only (≥ 500 kg)
- 9 seizure events/month on average
- Average of 12.3 metric tonnes/month
- Standard deviation: 7.7 metric tonnes
- High correlation with Mexican homicide rates
- Variation is mostly idiosyncratic (Colombian enforcement operations)



Empirical Strategy

Challenge: Violence is endogenous

- Reverse causality: dropouts may themselves affect violence
- Omitted variable bias: unobserved factors drive both

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- Instrument: $\underbrace{d_i}_{\text{distance to Pacific}} \times \underbrace{c_{j,t}}_{\text{Colombian seizures}}$
- Shift-share design, following Dube and Vargas [2013]

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Identifying assumption: Colombian seizures $\times$ Pacific distance affect dropout *only* through changes in local homicide rates

The instrument: two components

1. Colombian cocaine seizures ($c_{j,t}$): supply shocks

- Large Colombian seizures reduce cocaine supply
- Short-run demand is inelastic $\Rightarrow$ retail prices rise
- Mexican DTOs compete violently for control of trafficking routes
- Driven by Colombian enforcement operations: orthogonal to Mexico

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2. Distance to Pacific coast (d_i): exposure

- $\approx 70\%$ of Colombian cocaine enters Mexico via Pacific routes
- Closer to Pacific = more important within the trafficking network
- Time-invariant: identification from differential exposure to common shocks
- Consistent with Figure: highest homicide rates along Pacific corridor

First stage:

$$\ln(1 + h_{i,j,t}) = \gamma_1 + \gamma_2 [d_i \times c_{j,t}] + \mathbf{X}'_{i,j,t} \Gamma + \mathbf{W}'_{s,i,j,t} \Theta + \delta_{j,t} + m_s + \nu_{i,j,t}$$

Model

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Second stage:

$$y_{s,i,j,t} = \beta_1 + \beta_2 \ln(\widehat{1 + h_{i,j,t}}) + \mathbf{X}'_{i,j,t} \Gamma + \mathbf{W}'_{s,i,j,t} \Theta + \delta_{j,t} + m_s + \varepsilon_{s,i,j,t}$$

where:

- $y_{s,i,j,t} = 1$ if student s drops out in municipality i , year j , month t
- $\mathbf{X}_{i,j,t}$ = municipal controls (employment, income, population, lags of violence)
- $\mathbf{W}_{s,i,j,t}$ = individual/household controls
- m_s = individual fixed effects; $\delta_{j,t}$ = month-by-year fixed effects
- Standard errors clustered at the individual level

Instrument validity: uncorrelated with Mexican enforcement

	Marijuana (1)	Cocaine (2)	Heroin (3)	Meth (4)	Opium (5)	Total (6)
Colombian cocaine seizures (metric tonnes)	−4.220 (3.448)	−0.005 (0.017)	0.001* (0.001)	−0.034 (0.053)	−0.004 (0.003)	−4.262 (3.467)
Observations	12	12	12	12	12	12

Colombian seizures have **no significant correlation** with Mexican drug interdiction efforts across any substance (2007–2018). Annual data; 12 observations.

Instrument validity: uncorrelated with economic outcomes

	Municipal Outcomes			Individual Outcomes			National Outcome
	Employment Rate (1)	Avg Income (2)	Avg Weekly Hours (3)	HH Adult Employment (4)	HH Income (5)	HH Adult Hours (6)	Real GDP (7)
Colombian seizures (metric tonnes)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	−94.064 (3030.918)
Observations	83,584	83,808	83,808	2,526,299	2,582,435	2,352,527	72

Colombian seizures are uncorrelated with municipal employment rates, average income, weekly hours, household income, or national GDP.

Results

Main results

	OLS (1)	TWFE (2)	Minimal controls (3)	Primary specification (4)	War on drugs 2007-2012 (5)	Interim 2013-2016 (6)	Resurgence 2017-2024 (7)
<i>Outcome: Dropout</i>							
Homicides per 10,000	-0.003*** (0.000)	0.001 (0.001)	0.044 (0.084)	0.067 (0.087)	-0.197 (0.565)	-0.035 (0.139)	0.307** (0.137)
Mean of school enrollment	0.842	0.843	0.843	0.843	0.821	0.857	0.860
Kleibergen-Paap F-stat	—	—	164.06	154.75	3.96	64.02	65.44
Observations	2,032,058	1,857,143	1,973,879	1,857,143	742,496	507,571	578,125

Three key findings from baseline estimates

1. **OLS and TWFE** yield small, negative estimates — likely reflecting selection bias and time-invariant confounders
2. **Pooled IV**: positive point estimates — reversal from OLS suggests endogeneity was masking a positive effect; but pooled estimate is imprecise
3. **Time heterogeneity** (the central finding):
 - War on Drugs (2007–12): negative, imprecise; weak first stage
 - Interim Period (2013–16): negative, tight around zero; strong first stage
 - **Resurgence (2017–24): large, positive, significant at 5%**

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Kleibergen–Paap F-stat (pooled): 154.75 $\gg$ conventional thresholds

Resurgence period: effects concentrated in older males

<i>Outcome: Dropout</i>	Secondary school	High school	Male	Female
	(1)	(2)	(3)	(4)
Homicides per 10,000	0.147 (0.172)	0.413* (0.219)	0.495** (0.235)	0.138 (0.161)
Mean of school enrollment	0.962	0.784	0.842	0.880
Kleibergen-Paap F-stat	20.18	35.73	27.39	38.90
Observations	239,434	328,964	297,187	280,938

Magnitude of the Resurgence effect

A one-standard deviation increase in the municipal homicide rate (0.87 per 10,000):

- **All students (12–18):** increases dropout probability by ≈ 17.5 pp
 - Mean dropout rate $\approx 7.4\%$
- **High school students (15–18):** ≈ 23.5 pp
 - Mean $\approx 9.5\%$
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Results are **robust** to quarterly aggregation (Appendix Table)

Mechanism

Mechanism: DTO recruitment, not safety fears

If the mechanism were fear of **physical danger**, we would expect:

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Consistent with DTO recruitment:

- DTOs recruit older adolescent males as enforcers, distributors, operatives
- Younger teens serve in peripheral roles (lookouts) compatible with school
- Older teens accumulate “criminal capital” → full-time criminal careers [Sviatschi, 2022]
- DTO recruitment overwhelmingly targets young men

Why the Resurgence period specifically?

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AMLO's “abrazos, no balazos” (hugs, not bullets):

- Explicitly non-confrontational stance toward organized crime
- Reduced perceived risk of imprisonment or death from cartel membership
- Implicitly reduced the social stigma of participating in the drug trade
- Under Calderón: intense crackdowns $\Rightarrow$ high mortality risk of cartel membership
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Mexico's National Registry of Missing Persons:

- 3,000 missing in 2007 $\rightarrow$ 11,000 in 2012 $\rightarrow$ 35,600 in 2024
- Age-gender distribution: males peak at 15–19 and remain elevated through middle age
- Precisely the age range where dropout effects are largest

Robustness

Robustness checks

	Full Sample (1)	War on drugs (2)	Interim (3)	Resurgence (4)
Main specification	0.067 (0.087)	-0.197 (0.565)	-0.035 (0.139)	0.307** (0.137)
No 5 quarter restriction	0.083 (0.095)	-0.042 (0.538)	-0.033 (0.138)	0.383* (0.205)
Population threshold: 10,000	0.081 (0.095)	-4.721 (72.912)	-0.029 (0.130)	0.308** (0.134)
All cocaine seizures	0.009 (0.115)	-0.232 (0.716)	-0.008 (0.125)	0.175 (0.262)
Drop COVID (2020)	0.078 (0.087)	-0.197 (0.565)	-0.035 (0.139)	0.346** (0.139)
Outcome: School enrollment	-0.175 (0.123)	-0.603 (0.829)	0.166 (0.192)	-0.458** (0.198)

Robustness: summary

Positive Resurgence estimate is robust to:

- Including survey attriters (removing 5-quarter balance requirement)
- Lowering population threshold to 10,000
- Using all Colombian cocaine seizures (not just large coastal events)
- Dropping 2020 (COVID)
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Violence does **not** predict survey attrition:

- Small, statistically insignificant coefficients across all periods
- Only exception: slightly negative during the Interim period

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Policy implication: Scholarship programs targeting high school males in high-violence areas may reduce recruitment incentives of criminal organizations

References

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